Aman Verma

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WORK EXPERIENCE

Software Development Engineer Intern Nov 2025 - July 2026

LexoLegal *Remote*

- Architected 38 asynchronous REST API endpoints in **FastAPI** and **SQLAlchemy 2.0** across 12 relational tables behind **Nginx**, implementing multi-tier JWT authentication, RFC 9457 standardized problem details, and **Alembic** migrations (1,024 pytest suites).
- Deployed 'Munshi', an autonomous WhatsApp legal assistant in **Pydantic AI** and **FastAPI**, orchestrating **Model Context Protocol (MCP)** clients across Google Drive and Calendar servers to automate brief discovery and hearing scheduling with **sub-2s latency**.
- Hardened multi-tenant agent execution by isolating authenticated identity inside **Pydantic AI** AgentDeps (eliminating prompt injection vectors) and prevented worker race conditions using atomic **Redis** distributed locks with 10s **Celery** countdown requeues (90-retry ceiling).
- Automated an idempotent 12-hour case refresh sweep using **Celery** and **Redis** with a two-stage notification pipeline (dispatch dedup + deliver channel ledger) guaranteeing zero duplicate alerts.
- Reverse-engineered eCourts' proprietary mobile API protocol in **Python**, implementing **AES-128-CBC** cryptography with dynamic RIV salt generation and GIV lookup tables to ingest case data across 25 High Courts and 743 District Courts into **PostgreSQL**.

PROJECTS

ncert-rag-evals | *Python 3.14, Jupyter Notebooks, uv, SQLite FTS5, ChromaDB, Pytest* GitHub

- Constructed an empirical retrieval benchmark across 17 STEM textbooks in **Python** and **Jupyter Notebooks**, implementing custom hybrid search from scratch in **SQLite FTS5** and **ChromaDB** instead of high-level LlamaIndex abstractions to profile 8 retrieval strategies on raw CPU latency.
- Proved that fusing BM25 with dense embeddings via Reciprocal Rank Fusion (RRF) achieved **86.9% Recall@5 in 8.9ms on CPU**, matching LLM query expansion while cutting latency by **99.8%**.
- Engineered a document parser indexing 4,429 section-aware chunks into **SQLite FTS5** and **ChromaDB**; validated statistical significance via paired McNemar tests (**p = 0.0021**) across 92 unit tests in **Pytest**.

doubtless | *Pydantic AI, FastAPI, Multimodal RAG, Groq Whisper, Celery, Redis, Logfire* GitHub

- Engineered an interactive lecture doubt-solver and AI study workspace in **Pydantic AI** and **FastAPI**, fusing blackboard video frames with 45s playback dialogue windows to stream real-time solutions for visual STEM queries via SSE.
- Designed an asynchronous 4-stage media ingestion pipeline (PipelineRunner) in **Celery** and **Redis** that transcodes HLS streams, slices 16kHz mono audio (<15MB/hr), and transcribes lectures in <10s via **Groq Whisper-Large-V3**.
- Architected the async data layer with thread-safe **SQLite WAL** repositories and **Redis** to serve concurrent agent queries; containerized services and instrumented distributed tracing across agent runs via **OpenTelemetry** and **Logfire** (151 tests).

vectorless-rag | *Python, Jupyter Notebooks, Matplotlib, Pydantic AI, Ollama, ChromaDB, PyMuPDF* GitHub

- Evaluated dense vector search against LLM-based page indexing on benchmark documents using **Pydantic AI**, configuring **Ollama** as a local testbed alongside cloud LLMs across 18 benchmark queries.
- Benchmarked retrieval trade-offs in **Jupyter Notebooks** using **Matplotlib**: analyzed query failure cases and plotted the **2.7x token cost** and **2x latency** tradeoff (while verifying **100% retrieval hit rate** and **0.85 citation precision**).
- Built a zero-framework document ingestion engine in **Python** using **PyMuPDF** and **ChromaDB**, streaming page-aligned document windows with bounded memory (**<200MB**) to eliminate context boundary truncation and cut extraction latency by **40%**.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Bash, DSA, OOP, DBMS, Operating Systems

AI & Agent Frameworks: Pydantic AI, Model Context Protocol (MCP), LangGraph, LangChain, LlamaIndex, Ollama, Google ADK

Backend Frameworks: FastAPI, Celery, Asyncio, SQLAlchemy, Alembic, Nginx, REST APIs

Databases & Vector Stores: PostgreSQL, Redis, ChromaDB, SQLite (WAL)

Web Scraping & Automation: Selectolax, BeautifulSoup4, httpx, Playwright, Selenium, curl-cffi

Evals & Observability: OpenTelemetry, Langfuse, Logfire, Ragas, Pytest, Matplotlib, Jupyter Notebooks

Developer Tools & DevOps: Docker, Git, GitHub Actions, Postman, Linux, uv, Ruff

LEADERSHIP & ACTIVITIES

- Delivered 3 technical sessions as a speaker on 'Generative AI Unlocked: LLM Basics' and 'Level Up Your Career with AI', training 150+ college students on foundational LLM concepts and modern AI developer tools.
- Coordinated placement drives and corporate outreach at CTPC (Central Training & Placement Cell), managing logistics and interview scheduling for recruiters and 100+ candidates.
- Organized Smart India Hackathon internal rounds across 2 editions; coordinated problem statement curation, mentoring, and judging logistics for 20+ teams and 120+ participants.

EDUCATION

Veer Bahadur Singh Purvanchal University 2023 - 2027

Bachelor of Technology in Computer Science & Engineering *Jaunpur, India*